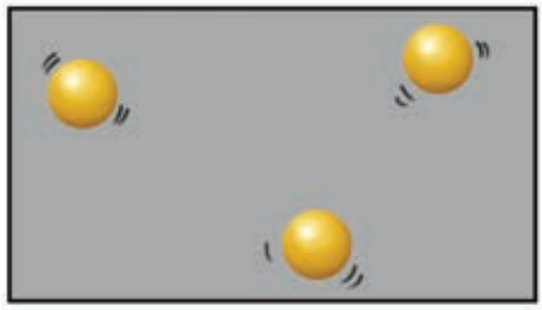
water vapour, if you heat it sufficiently.

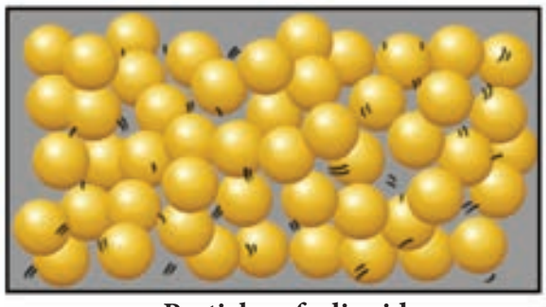
However, when you put a wet cloth to dry, the water evaporates into air, leaving the clothes dry.

That is there are two types of vaporization: boiling and evaporation, the first one is by heating and the second type of vapourization is natural.

Boiling is the process of conversion of a liquid into vapours on heating. In gaseous state, only the arrangement of molecules changes, there is no change in their chemical composition. So, boiling is a physical change.

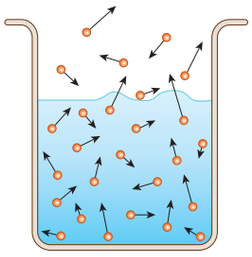


**Particles of a gas**



**Particles of a liquid**

Upon heating a liquid, the particles gain energy and vibrate more vigorously. When the particles possess enough energy, they overcome the strong forces of attraction between one another. The particles break free from one another and move randomly. For example, when liquid water is heated to 100°C, it boils to become steam. Boiling occurs when the boiling point is reached. The liquid changes to its gaseous state.

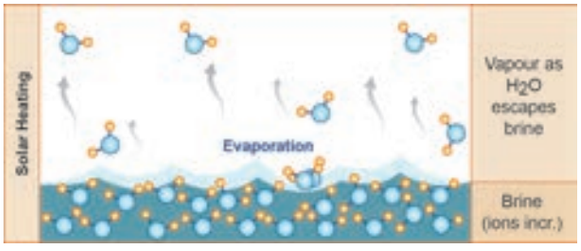


## Evaporation

Take a glass of water. All the water molecules are moving here and there at different velocities (shown as arrows of different lengths). Some of the molecules, especially at the surface, could be moving in a direction away from the liquid, and have adequate energy to overcome the attractive force (surface tension) of the liquid, then that molecule will escape into the air. Thus slowly and steadily the water molecules escape, or said to evaporate, and the water level in the glass decreases as the time passes. Note that the temperature of the water did not rise to the level of boiling point of water. Nor were there any bubbles formed like boiling.

Evaporation is the technique used to separate dissolved solids from a solid-liquid mixture. This is the technique used to extract salt from sea water in salt pans. Shallow level of sea water is impounded. Slowly the water evaporates due to action of Sun. Ultimately salt deposits over the ground we can understand. Evaporation makes use of the fact that the solvent in a solution can vapourise at any temperature, leaving behind a residue of the solid that was dissolved in the liquid.

From drying clothes to drying fish, evaporation is used.




**Factors affecting the rate of evaporation**



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